The Nurse Scheduling Problem

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1 Description of the Problem

The problem in question is to create an optimal shift schedule for nurses working in a hospital during a given period of days, respecting some constraints:

(1) Each shift has a minimum number of nurses necessary for the normal functioning of the hospital.
(2) Each shift requires at least one nurse with managerial qualifications.
(3) A nurse's daily schedule has to be continuous, meaning that there are no gaps in between assigned shifts.
(4) Each nurse has a minimum and maximum daily workload.
(5) The total number of shifts that can be assigned to a nurse for the whole period is limited.
(6) A nurse cannot be assigned to shifts when it's known they're unavailable to work.
(7) All nurses' shift preferences, known beforehand, have to be taken into account, being an optimal solution one in which the most nurses are happy with their schedule.

2 Description of the Encoding

This problem can be modeled as a MaxSAT problem. To solve it we first need to define variables which can then be used to encode the constraints above. This encoding can then be fed into a MaxSAT solver and an optimal solution will be generated. Let us define the variables of the problem.

2.1 Definition of Variables

- Let D be a positive integer representing the number of days.
- Let S be a positive integer representing the number of shifts per day, all with the same duration, that cover the full 24 hours. For example if $S = 4$, each shift will have 6 hours.
- Let $R_{s,d}$ define the minimum number of nurses required for shift s $(1 \leq s \leq S)$ at day d $(1 \leq d \leq D)$.
- Let $N = \{N_1, N_2, ..., N_n\}$ be a set of n positive integers representing the available nurses to be assigned to work in the shifts.
- Let $M \subseteq N$ define the subset of nurses with managerial qualifications.
- Let k_{min} and k_{max} define the minimum and maximum number of shifts for nurse N_k $(1 \leq k \leq n)$ in any given day.
- Let H_k define the maximum total number of shifts that can be assigned to nurse N_k for the whole period.
- Let $P_{k,s,d}$ be a set of $n \times S \times D$ integers representing the cost of assigning nurse N_k to shift s in day d. If $P_{k,s,d} = -1$, nurse N_k is not available for shift s in day d. All other values in P are positive.
- Let $X = \{x_1, x_2, ..., x_i\}$, for any $i \leq n \times S \times D$, be a set of variables, each taking a value in $\{0, 1\}$ such that $x_i = 1$ iff the nurse is assigned to that shift. What nurse is assigned to what shift in which day is clarified below.
- Let there be within X k subsets X_k of $S \times D$ variables, each associated with nurse N_k. For example, X_1 is associated with nurse N_1 and contains the first $S \times D$ variables, X_2 is associated with nurse N_2 and contains the next $S \times D$ variables, and so on.
- Let there be within each X_k D subsets $X_{k,d}$ of S variables, each associated with nurse N_k and day d. For example, $X_{1,1}$ represents $d = 1$ of N_1 and contains the first S variables, $X_{1,2}$ represents $d = 2$ of N_1 and contains the next S variables, and so one.
- Within each $X_{k,d}$, there are S variables, each representing the participation of nurse N_k for day d in each shift. For example, $X_{1,1,1} = x_1$ and encodes the participation of nurse N_1 in shift $s = 1$ in day $d = 1$, $X_{1,1,2} = x_2$ and encodes the participation of nurse N_2 in shift $s = 1$ in day $d = 1$, and so on. We can essentially represent x_i as $X_{k,d,s}$ and vice-versa. An example will be given below for better understanding.
- Let $L_{d,s}$ be a subset of X that contains all variables x_i associated with shift s of day d.

Let $D = 2, S = 2$ and $N = \{N_1, N_2, N_3\}$ such that $X = \{x_1, x_2, ...x_{12}\}$.

Nurse 1	Shift 1	Shift 2
Day 1	x_1	x_2
Day 2	x_3	x_4

Nurse 2	Shift 1	Shift 2
Day 1	x_5	x_6
Day 2	x_7	x_8

Nurse 3	Shift 1	Shift 2
Day 1	x_9	x_{10}
Day 2	x_{11}	x_{12}

Any $X_{k,s,d}$ can be represented as x_i:

$$i = (S \cdot D)(k-1) + S(d-1) + s$$

On the contrary, for $i = 7$, we know that $x_i \in X_2$ due to the size of any X_k being $D * S$, and further we know that $x_i \in X_{2,2}$ due to the

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size of any $X_{k,d}$ being S. Finally, we can use the previous equation to find that $x_7 = X_{2,2,1}$.

Thus, $x_i = X_{k,s,d} = 1$ iff nurse N_k is assigned to shift s in day d. For instance, $x_9 = 1$ iff nurse N_3 is assigned to the shift $s = 1, d = 1$. Within our program this set X is cleanly managed by the PySAT IDPool class.

2.2 Modeling Constraints

Having defined the necessary variables, we can now describe the encoding we chose to model the initial constraints.

$$\sum_{x_j \in L_{d,s}} x_j \geq R_{s,d}, \quad \forall d \in [1, D], s \in [1, S] \tag{1}$$

$$\sum_{x_j \in L_{s,d}} x_j \geq 1, \quad \forall k, s, d : s \in [1, S], d \in [1, D], N_k \in M \tag{2}$$

$$\begin{gathered}(\neg X_{k,s,d} \vee X_{k,s+1,d} \vee \neg X_{k,v,d}), \\ \forall k, s, d, v : k \in [1, n], s \in [1, S-1], d \in [1, D], v \in [s+2, S]\end{gathered} \tag{3}$$

$$\begin{gathered}\sum_{x_j \in X_{k,d}} x_j \geq k_{\min}, \quad \forall k \in [1, n], d \in [1, D] \\ \sum_{x_j \in X_{k,d}} x_j \leq k_{\max}, \quad \forall k \in [1, n], d \in [1, D]\end{gathered} \tag{4}$$

$$\sum_{x_j \in X_k} x_j \leq H_k, \quad \forall k \in [1, n] \tag{5}$$

$$(\neg X_{k,d,s}), \quad \forall k, d, s : P_{k,s,d} < 0 \tag{6}$$

We guarantee an optimal solution will be found by adding the following soft clauses:

$$(\neg X_{k,d,s}) \text{ with weight } P_{k,s,d}, \quad \forall k, s, d : P_{k,s,d} \geq 0 \tag{7}$$

3 Algorithm and Configuration

With the encoding defined and coded into Python, we fed the formula into the RC2 MaxSAT solver [2] available in PySAT, using the CaDiCaL 1.9.5 oracle [1] underneath. We chose *cadical*195 simply because it far outperformed every other option we tried. All cardinality constraints use a Modulo Totalizer encoding [3].

4 Installing and running

To solve this project we used Python. The only dependency used in the project was PySAT which can be installed by running the following command: `pip3 install pysat`. To run the program, simply execute:

```
python3 project1.py < input.txt > output.txt
```

Note that `input.txt` and `output.txt` should be replaced with your actual file names.

References

[1] Armin Biere, Tobias Faller, Katalin Fazekas, Mathias Fleury, Nils Froleyks, and Florian Pollitt. 2024. CaDiCaL 2.0. In *Computer Aided Verification - 36th International Conference, CAV 2024, Montreal, QC, Canada, July 24-27, 2024, Proceedings, Part I (Lecture Notes in Computer Science, Vol. 14681)*, Arie Gurfinkel and Vijay Ganesh (Eds.). Springer, 133–152. https://doi.org/10.1007/978-3-031-65627-9_7

[2] Alexey Ignatiev, Antonio Morgado, and Joao Marques-Silva. 2019. RC2: an Efficient MaxSAT Solver. *Journal on Satisfiability, Boolean Modelling and Computation* 11, 1 (2019), 53–64. https://doi.org/10.3233/SAT190116

[3] Toru Ogawa, Yangyang Liu, Ryuzo Hasegawa, Miyuki Koshimura, and Hiroshi Fujita. 2013. Modulo Based CNF Encoding of Cardinality Constraints and Its Application to MaxSAT Solvers. In *2013 IEEE 25th International Conference on Tools with Artificial Intelligence*. 9–17. https://doi.org/10.1109/ICTAI.2013.13